

--	--	--	--	--	--	--	--

B.Tech. Degree V Semester Examination November 2017

IT 15-1504 DESIGN AND ANALYSIS OF ALGORITHMS (2015 Scheme)

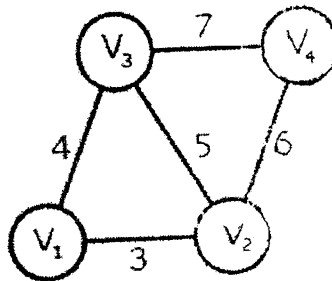
Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) What do you mean by asymptotic growth rate?
 (b) Differentiate divide and conquer and greedy design techniques.
 (c) Which of the given options provides the increasing order of asymptotic complexity of functions f_1 , f_2 , f_3 and f_4 ?
- $f_1(n) = 2^n$
 $f_2(n) = n^{(3/2)}$
 $f_3(n) = n \log n$
 $f_4(n) = n^{(\log n)}$
- (d) Compare various sorting algorithms based on their best, average and worst case.
 (e) List the properties of red black trees.
 (f) An undirected graph $G(V, E)$ contains n ($n > 2$) nodes named $v_1, v_2, \dots, v_n$. Two nodes v_i, v_j are connected if and only if $0 < |i - j| \leq 2$. Each edge (v_i, v_j) is assigned a weight $i + j$. A sample graph with $n = 4$ is shown below. What will be the cost of the minimum spanning tree (MST) of such a graph with n nodes?



- (i) $\frac{1}{12}(11n^2 - 5n)$
 (ii) $n^2 - n + 1$
 (iii) $6n - 11$
 (iv) $2n + 1$
- (g) Define an optimal binary search tree.
 (h) Compare two graph traversal techniques.
 (i) What is the need for classifying problems into classes based on their complexity?
 (j) What do you mean by approximation algorithm?

(P.T.O.)



PART B

(4 × 10 = 40)

- II. Discuss the asymptotic notations used in the analysis of algorithms. (10)

OR

- III. Solve the following recurrence equations:

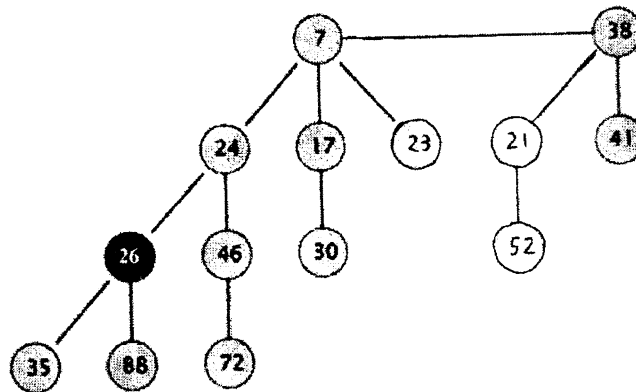
(a) $T(n) = T(n/2) + 2^n$ (using Master's theorem). (5)

(b) $T(n) = 8T(n/2) + n^2$ (using Recursion Tree method). (5)

- IV. Analyze the average case and worst case time complexity of quick sort algorithm. (10)

OR

- V. (a) Perform decrease key operation on the following Fibonacci heap in the given order. (5)



- (i) Decrease key of 46 to 29.
 (ii) Decrease key of 29 to 15.
 (iii) Decrease key of 35 to 5.
- (b) Insert 2, 1, 4, 5, 9, 3, 6, 7 in a Red Black tree and delete 4, 6. (5)
- VI. (a) Describe transitive closure of a binary relation. (5)
 (b) Define strongly connected components of a directed graph. Also write the algorithm to find the strongly connected components. (5)

OR

- VII. Discuss Floyd's algorithm to find all pair shortest path using suitable example. (10)
- VIII. What is travelling salesman problem? Explain two possible strategies used for travelling salesman problem. (10)

OR

- IX. (a) Illustrate the idea behind Wigderson's graph coloring algorithm. (5)
 (b) Differentiate NP hard and NP complete class of problems. (5)
